

Reactive Blocks

Q32

STATE (LEGACY) BLOCK LOGIC

INTERVIEW QUESTION

Question. Can a legacy `$:` statement block execute multi-line logic, blocks, or external function calls? Provide a conceptual example.

ANSWER. Yes. Because `$:` is simply valid JavaScript syntax (a labeled statement), it can precede any statement. By combining it with a curly brace block `{ ... }`, you can execute multi-line logic, conditional `if/else` statements, or trigger external function calls. The compiler will scan the entire block and run it whenever any of its internal top-level dependencies are updated.

App.svelte

```
<script>
  let score = 85;
  let grade = 'F';

  // The $: label applies to the entire block.
  // When 'score' changes, the entire block runs again.
  $: {
    if (score ≥ 90) {
      grade = 'A';
      triggerConfetti(); // External function
    } else if (score ≥ 80) {
      grade = 'B';
    } else {
      grade = 'F';
      logFailure(score);
    }
  }
}
```

</script>

- ① **Summary.** By attaching the `$:` label to a block `{ ... }`, you can wrap complex, multi-line conditional logic and side-effects into a single reactive bundle that re-evaluates automatically based on its dependencies.

